ServiceNow Universal MCP

LLM-Agnostic Integration for the Enterprise

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Connect any LLM to ServiceNow. No vendor lock-in. One protocol.

26 Tools · 11 ServiceNow Modules · 5 LLM Providers · 2 Transport Modes · AGPL-3.0

Executive Summary

In May 2026, a LinkedIn post demonstrated MCP integration connecting Claude to ServiceNow. The vision was compelling: *the future of enterprise software is conversational, not form-driven.*

The limitation was equally clear: that implementation was **locked to Claude Desktop and the Anthropic ecosystem.**

ServiceNow Universal MCP removes this constraint. It provides the same conversational interface to ServiceNow — across OpenAI, Anthropic, DeepSeek, Ollama, OpenRouter, and any OpenAI-compatible API. One protocol, zero vendor dependency.

The Problem

Enterprise ServiceNow teams spend a measurable portion of their day navigating ITSM forms rather than solving problems:

- **5 to 7 minutes** to create a single incident — field-by-field navigation through category trees, assignment group lookups, and priority matrices

- **3 to 4 separate screens** to track a service request from submission through fulfillment
- **Context fragmentation** — CMDB searches, knowledge base lookups, and performance reports live in disconnected modules
- **Tool switching tax** — each context switch between modules costs approximately 23 seconds of cognitive recovery time

The accumulated cost across a team of ten engineers exceeds **400 hours per month** in navigation overhead alone.

The Solution

ServiceNow Universal MCP collapses all module interaction into a single conversational interface. An engineer types a natural-language request and the server executes it across the appropriate ServiceNow APIs.

Capability	Claude MCP	Universal MCP
Claude (Anthropic)	Yes	Yes
OpenAI GPT-4o	No	Yes
DeepSeek	No	Yes
Ollama (local models)	No	Yes
OpenRouter (150+ models)	No	Yes
Any OpenAI-compatible API	No	Yes
STDIO transport	Yes	Yes
HTTP transport (web clients)	No	Yes
ServiceNow modules	~5	**11**
Pydantic input validation	No	Yes
Structured JSON logging	No	Yes
Docker support	No	Yes

Module Coverage

Incident Management — 5 tools Create, search, update, and aggregate statistics. Filter by state, priority, assignment group, and assignee. Paginated queries with accurate total counts via the stats API.

Change Management — 3 tools Create Change Requests with risk classification. Approve or reject with audit commentary. Filter by type (normal, standard, emergency) and approval status.

Problem Management — 3 tools Create problem records for root-cause analysis. Link incidents to problems for traceability. Filter by state and assignment group.

Service Catalog & Requests — 4 tools Browse 197 catalog items. Submit requests through the Cart API. Track status across REQ, RITM, and SCTASK tables. View pending approvals.

CMDB — Configuration Management — 3 tools Search configuration items by name, class, and environment. Map parent-child dependency relationships. Run health diagnostics: duplicate detection, orphan identification, and stale CI reporting (90-day threshold).

Knowledge Base — 1 tool Full-text search across published articles with category filtering.

Reporting & Analytics — 4 metrics SLA breach counts, mean time to resolution, group workload distribution, and overdue incident trends with percentage calculations.

Infrastructure Modules — 5 tools Workflow inventory, REST integration catalog, business rule auditing per table, user lookup, and group membership enumeration.

Architecture

```
┌──────────────────────────────────────────────────────────────┐
│   LLM: GPT-4o / Claude Sonnet / DeepSeek / Ollama / Router   │
│                         │ tool call                            │
├──────────────────────────────────────────────────────────────┤
│                      UniversalMCPServer                        │
│  ┌───────────────┐  ┌───────────────┐  ┌───────────────┐     │
│  │ MCP Protocol  │  │  LLM Adapter  │  │ SN REST Client│     │
│  │ (STDIO/HTTP)  │  │ (abstraction) │  │ (httpx+retry) │     │
│  └───────────────┘  └───────────────┘  └───────────────┘     │
│                         │                                      │
│  ┌──────────────────────────────────────────────────────┐   │
│  │  Pydantic Validation Layer — 26 argument models      │   │
│  └──────────────────────────────────────────────────────┘   │
│                         │                                      │
│  ┌──────────────────────────────────────────────────────┐   │
│  │  ToolHandlers (dispatcher) — 26 handlers             │   │
│  │  incident_create │ change_approve │ cmdb_search      │   │
│  │  problem_link    │ catalog_list   │ kb_search        │   │
│  └──────────────────────────────────────────────────────┘   │
│                                                                │
└──────────────────────────────────────────────────────────────┘
```

```
                ServiceNow Instance
     /api/now/table/*    /api/now/stats/*
     /api/sn_sc/servicecatalog/cart
```

Interaction Examples

> *"Create a P1 incident for the production database outage. Assign to the DBA team."*
>
> The server resolves the DBA group, creates the incident record, and returns the confirmed incident number.
>
> *"Which changes are scheduled for this weekend, and what is their approval status?"*
>
> The server queries change_request for the specified window and returns a filtered list with approval states.
>
> *"Which team carries the highest overdue incident load this month?"*
>
> The server aggregates active incidents by assignment group, cross-references overdue counts, and returns the ranked distribution.
>
> *"Find all production servers running Windows Server 2019."*
>
> The server searches cmdb_ci with environment and class filters, returning matching configuration items.

Deployment

Installation

```
git clone https://github.com/vladarchitectservicenow-oss/servicenow-uni
cd servicenow-universal-mcp
pip install -e ".[all]"
```

Configuration

```
cp .env.example .env
# Required: SNOW_INSTANCE_URL, SNOW_USERNAME, SNOW_PASSWORD
# Any one LLM key:
#   OPENAI_API_KEY        → GPT-4o
```

```
#   ANTHROPIC_API_KEY     → Claude
#   DEEPSEEK_API_KEY      → DeepSeek
#   OLLAMA_HOST           → Local model
#   OPENROUTER_API_KEY    → 150+ models
```

The server auto-discovers the available provider. Configure one key; skip the rest.

Execution

```
sn-mcp --stdio              # MCP clients (Claude Desktop, Continue, Cl
sn-mcp --http --port 8000   # Web clients (Open WebUI, custom frontends
```

Docker

```
docker compose up -d        # Server available at http://localhost:8000
```

Supported Providers

Provider	Representative Models	Configuration Variable
OpenAI	GPT-4o, GPT-4-turbo	`OPENAI_API_KEY`
Anthropic	Claude Opus, Sonnet, Haiku	`ANTHROPIC_API_KEY`
DeepSeek	deepseek-chat, deepseek-reasoner	`DEEPSEEK_API_KEY`
Ollama	Llama 3, Qwen 2.5, Mistral, Gemma	`OLLAMA_HOST`
OpenRouter	150+ models	`OPENROUTER_API_KEY`
Custom	Any OpenAI-compatible API	`base_url` + `api_key`

Verified Instance Metrics

Tested against a ServiceNow PDI running the Australia release:

- **197 catalog items** (147 active)
- **6 workflows** (published flows)
- **11 external integrations** — Azure AD, Slack, Jira, Okta, AWS, SAP, Datadog
- **2,784 configuration items** in CMDB
- **220 dependency relationships**
- **53 knowledge base articles**
- **79 Virtual Agent topics**
- **409 Performance Analytics indicators**

Estimated Impact

Metric	Before	After
Time to create an incident	5–7 minutes	~15 seconds
Screens per status check	3–4	1 (conversation)
Module context switches per task	2–3	0
Onboarding time for new tool	Hours (per module)	Minutes (one interface)
Daily navigation overhead per engineer	40–60 minutes	~5 minutes

For a team of ten engineers: **400+ hours recovered per month.**

Quality & Operations

- **46 automated tests** covering all 26 tool handlers — pytest with asyncio support
- **Pydantic validation** on every tool invocation — type checking, range constraints, required field enforcement
- **Structured JSON logging** — machine-parseable, request-level traceability
- **Retry with exponential backoff** — 3 attempts via tenacity for transient ServiceNow API failures
- **Multi-stage Docker build** — python:3.12-slim, non-root execution, health check endpoint
- **CI/CD pipeline** — lint (ruff + mypy), cross-version test matrix (3.10–3.12), Docker verification, artifact publication

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Roadmap

- Virtual Agent & AI Agent Studio module (Australia release components)
- Service Portal / Employee Center module
- Automated Test Framework (ATF) integration
- OpenTelemetry distributed tracing
- Aggregate API adoption for statistical queries
- Helm chart for Kubernetes deployment

Built by Vlady — ServiceNow Developer & AI Architect — May 2026

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